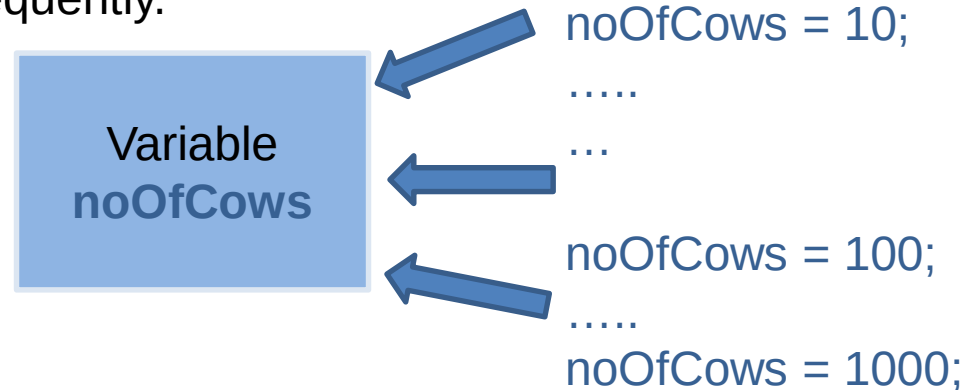


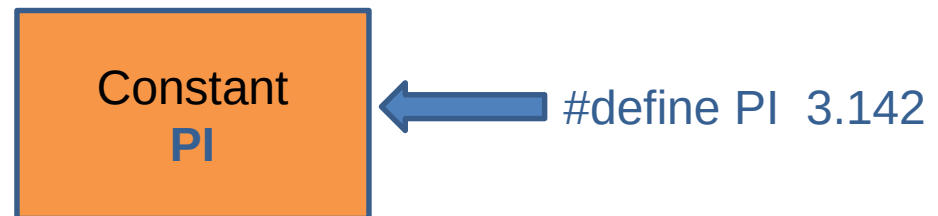
Lab 8: Programming basics II

- Variables and constants

A **variable** can be initialized to any value of its **type**, and assigned again and again with a different value subsequently.



A **constant** must be initialized to a (fixed) value and cannot be changed subsequently.



Basic Elements of Programming

OUTPUT

CALCULATIONS

VARIABLES

DECISIONS

CONSTANTS

REPETITION

INPUT

FUNCTIONS



A program is made up of the above basic elements.
This topic will examine ***variables*** and ***CONSTANTS***.

Review

- Variable declaration

Type **variable_name**; e.g. **int yearOfBirth;**

- printf()** : the library function that displays a message on the screen

printf(format string, arg1, arg2, ...)

- What is a format string? Format tags whose values are substitute by args.
- What are format tags?

- scanf()** : the lib function that reads the data typed in by user using keyboard

scanf(format string, &arg1, &arg2, ...)

- Store values into variables (i.e. locations pointed by args)

```
void main()
```

```
{
```

```
    int age;
```

```
    char gender;
```

```
    float salary;
```

```
    printf("Enter age and salary: ");
```

```
    scanf("%d%f", &age, &salary);
```

```
    printf("Enter gender (F/M): ");
```

```
    fflush(stdin); //clear the \r\n
```

```
    scanf("%c", &gender);
```

```
    printf("you entered => age = %d gender = %c salary = %.2f\n", age, gender, salary);
```

```
}
```

```
Enter age and salary: 18 2350.75
```

```
Enter gender (F/M): F
```

```
you entered => age = 18 gender = F salary = $2350.75
```

Integer or fraction number?

- Integer or fraction is more appropriate for storing the following data?

integer (int)	fraction (float)

1. Temperature in degrees Celsius
2. The population of a city
3. Your grade point average
4. A person's age in years
5. A person's weight in pounds
6. A person's height in meters
7. Number of miles traveled
8. Your locker number
9. Number of seconds left in a game
10. The sum of integers
11. The sum of double numbers

Constants

A **constant** is a symbolic name used to replace a (fixed) value that should not be modified later by the program.

//Declaring and Initialising Constants

//Declare and initialize constants before the **main()** function

//E.g. :

```
#include <stdio.h>
```

```
#define PI 3.142
```

```
#define SIZE 3
```

```
#define CAP 'A'
```

```
void main()
```

```
{
```

```
    printf("PI = %f \n", PI);
```

```
    printf("SIZE = %d \n", SIZE);
```

```
    printf("CAP = %c \n", CAP);
```

```
}
```

Increment/Decrement operators, ++ and --

```
#include <stdio.h>
```

```
void main()
```

```
{
```

```
    int c;
```

```
    c = 5;
```

```
    printf("%d \n", c);    // 5
```

```
    c++;                  //variable c is incremented by 1
```

```
    printf("%d \n\n", c);  // 6
```

```
    c = 5;
```

```
    printf("%d \n", c);    // 5
```

```
    c--;                  //variable c is decremented by 1
```

```
    printf("%d\n", c);     // 4
```

```
}
```

A2. Additional Exercise

Draw a flowchart & write the corresponding program to calculate the travel charge for a pizza delivery person at a rate of \$3.50 per kilometer.

Your program should interact with the user in the following manner:

TRAVEL CHARGE CALCULATOR

Enter beginning odometer readings=> 130.2

Enter ending odometer readings=> 138.1

You traveled 8.1 kilometers. At \$3.50 per kilometer,

Your travel charge is \$28.35